Mahesh Chitakoti

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Summary

Full Stack Engineer with 3+ years of software engineering experience building web applications with **React, Next.js, TypeScript, Node.js, and FastAPI**. Improved signup conversion by **18%** through onboarding experiments and reduced median API response time by **60%** through PostgreSQL indexing and Redis caching.

Technical Skills

Frontend Engineering: React.js, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, Tailwind CSS, Vite

Backend & APIs: Node.js, Express.js, NestJS, Python, FastAPI, AsyncIO, REST APIs, WebSockets, Microservices, Idempotent Webhooks

Databases & Caching: PostgreSQL, Redis (Caching, Task Queues), MongoDB, Compound Indexing, Connection Pooling

Cloud, DevOps & Quality: AWS (EC2, S3), Docker, GitHub Actions CI/CD, Jest, Postman API Testing, Linux, Nginx

Applied AI: RAG, Qdrant, pgvector, Gemini & OpenAI APIs

Experience

Independent Software Developer – Contract & Client Projects Apr 2026 – Present

Remote

- Built a **FastAPI** backend that handled client webhooks and used **Redis queues** to schedule data extraction and schema validation for multi-page PDF invoices.
- Deployed containerized services with **Docker on AWS EC2**, provisioned **S3** object storage, and automated linting and tests with **GitHub Actions**.
- Reduced average vector retrieval latency from **280 ms to under 185 ms** using **PostgreSQL with pgvector HNSW indexing** and semantic caching.

AI & Full Stack Developer – SahioNext Technology Pvt. Ltd. Feb 2025 – Apr 2026

Remote

- Built operational dashboards in **React.js and Redux Toolkit** to track token usage, rate limits, and pipeline bottlenecks, cutting incident triage from hours to minutes.
- Developed **React.js** interfaces and **FastAPI** services for Tulkka's audio evaluation workflows, using PostgreSQL and dispatching processing jobs to workers through Redis queues.
- Implemented full-duplex **WebSocket endpoints** for real-time speech evaluation, using client-side buffering and automatic reconnection to recover sessions on unstable networks.
- Integrated **Qdrant hybrid search** into the evaluation workflow to retrieve relevant content from **10,000+ lesson transcript chunks** at p95 latency under **300 ms**.

Full Stack Developer – The DevOps Team Oct 2023 – Jan 2025

Bangalore, India

- Built responsive onboarding and profile workflows in **React.js and Next.js**. A/B testing changes to the multi-step signup form increased registration conversion by **18%**.
- Optimized **Node.js and Express.js APIs** using Redis caching, PostgreSQL compound indexes, and connection pooling, reducing median response times from **800 ms to 320 ms**.
- Added **Jest** unit tests and **Postman** API tests for core authentication and search endpoints.

Projects

WorkBridge – Full-Stack Collaboration Platform *Sep 2026 – Present*

Next.js, React.js, TypeScript, Node.js, Express.js, FastAPI, PostgreSQL, pgvector, Redis, Tailwind CSS | Live Project

- Built real-time project boards and role-matching filters with **Next.js, React.js, and Tailwind CSS**. Used optimistic UI updates to show board changes immediately.
- Implemented a **Node.js/Express API gateway** for OAuth 2.0 flows, session handling, and role-based access control, with a separate **FastAPI** service for vector search.
- Benchmarked search endpoints with **k6** and sustained p95 latency under **150 ms** at 50 simulated virtual users after tuning PostgreSQL connection pooling and Redis session caching.

Certifications

Oracle Agentic AI Certified Foundations Associate – Oracle — Aug 2026
Oracle AI Vector Search Certified Professional – Oracle — Sep 2025
Machine Learning I – Columbia University — Aug 2025
Oracle Cloud Infrastructure Generative AI Professional – Oracle — Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU) — Chikkamagaluru, KA, India
Bachelor of Engineering — *Aug 2023*